



Python Programming - 2301CS404

Lab - 8

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01) WAP to find their union, intersection, and difference of the given two sets.

Sample Input:

A = {1, 2, 3, 4}
B = {3, 4, 5, 6}

Sample Output:

Union: {1, 2, 3, 4, 5, 6}
Intersection: {3, 4}
Difference (A-B): {1, 2}

```
In [9]: A = {1,2,3,4}
        B = {3,4,5,6}

        Union_Set= {}
        Intersection_Set = {}
        Difference_Set = {}

        Union_Set = A.union(B)
        print("Union:",Union_Set)

        Intersection_Set = A.intersection(B)
        print("Intersection:",Intersection_Set)

        Difference_Set = A.difference(B)
        print("Difference:",Difference_Set)
```

Union: {1, 2, 3, 4, 5, 6}

Intersection: {3, 4}

Difference: {1, 2}

02) WAP to Find elements common to all three given sets.

Sample Input:

A = {1, 2, 3, 4}

B = {2, 3, 5}

C = {2, 3, 6}

Sample Output:

{2, 3}

```
In [10]: A2 = {1,2,3,4}
          B2 = {2,3,5}
          C2 = {2,3,6}

          InterSectionSet2 = {}

          InterSectionSet2 = A2.intersection(B2)
          InterSectionSet2 = InterSectionSet2.intersection(C2)
          print("InterSection Set:",InterSectionSet2)
```

InterSection Set: {2, 3}

03) WAP to remove duplicate elements from a list while preserving unique values.

Sample Input:

[10, 20, 10, 30, 20, 40]

Sample Output:

{10, 20, 30, 40}

```
In [18]: li3 = [10,20,10,20,30,40]

          newSet = set(li3)
          print(newSet)
          print("New Set:",newSet.union(li3))
```

{40, 10, 20, 30}

New Set: {40, 10, 20, 30}

04) WAP to count how many elements are common between two sets.

Sample Input:

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
```

Sample Output:

2

```
In [26]: A4 = {1,2,3,4}
B4 = {3,4,5,6}
count4 = 0
set4 = {}
set4 = A4.intersection(B4)
for i in range(0,len(set4),1):
    count4 += 1
print("Common elements Count:",count4)
```

Common elements Count: 2

05) WAP to Merge two dictionaries. If a key exists in both dictionaries, add their values.**Sample Input:**

```
Dict1 = {1: 10, 2: 20, 3: 30}
Dict2 = {2: 40, 3: 50, 4: 60}
```

Sample Output:

```
{1: 10, 2: 60, 3: 80, 4: 60}
```

```
In [39]: Dict1 = {1: 10, 2: 20, 3: 30}
Dict2 = {2: 40, 3: 50, 4: 60}

# MergedDict = {}

for i,j in Dict1.items():
    if i in Dict2:
        Dict2[i] += j
    else:
        Dict2[i] = j

print(Dict2)
```

{2: 60, 3: 80, 4: 60, 1: 10}

06) WAP to Sort a dictionary in ascending order of its values.

Hint: in sorted(), use udf for key parameter

Sample Input:

```
{'Math': 70, 'Physics': 85, 'Chemistry': 60}
```

Sample Output:

```
{'Chemistry': 60, 'Math': 70, 'Physics': 85}
```

```
In [40]: Set6 = {'Math': 70, 'Physics': 85, 'Chemistry': 60}

new_dict6 = dict(sorted(Set6.items(), key = lambda i : i[1]))
print(new_dict6)
```

```
{'Chemistry': 60, 'Math': 70, 'Physics': 85}
```

07) WAP to find the keys having maximum and minimum values in a dictionary.**Sample Input:**

```
{'A': 50, 'B': 90, 'C': 30}
```

Sample Output:

```
Max Value Key: B
Min Value Key: C
```

```
In [69]: Dict7 = {'A': 50, 'B': 90, 'C': 30}
maximum = 0
minimum = 0
maxi = ""
mini = ""
# ne = max(Dict7, key=Dict7.get)
# print(ne)

for i,j in Dict7.items():
    if maximum<j:
        maximum = j
        maxi = i

for i,j in Dict7.items():
    if minimum>j:
        minimum = j
        mini = i

print(maxi)
print(mini)
```

B

08) WAP to create a dictionary with numbers from 1 to N as keys and their squares as values.**Sample Input:**

```
N = 5
```

Sample Output:

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

```
In [63]: n = int(input("Enter The Number: "))

dict8 = {}
for i in range (1,n+1,1):
    dict8[i] = i**2

print(dict8)
```

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

09) WAP to count the frequency of each character in a string using a dictionary.**Sample Input:**

```
programming
```

Sample Output:

```
{'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}
```

```
In [80]: s = "programming"
dict9 = {}

for i in s:
    if i in dict9:
        dict9[i] += 1
    else:
        dict9[i] = 1

print(dict9)
```

```
{'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}
```

10) WAP to swap keys and values of a dictionary.**Sample Input:**

```
{1: 'A', 2: 'B', 3: 'C'}
```

Sample Output:

```
{'A': 1, 'B': 2, 'C': 3}
```

```
In [89]: Dict10 = {1: "A",2:"B", 3:"C",4:"D"}
newDict10 = {}

for i,j in Dict10.items():
    newDict10[j] = i
print(newDict10)
```

```
{'A': 1, 'B': 2, 'C': 3, 'D': 4}
```

11) WAP to find the student with the highest total marks, given student names and their marks.

Sample Input:

```
{'Amit': [70, 80, 90], 'Riya': [85, 75, 80], 'Neha': [60, 70, 65]}
```

Sample Output:

Topper: Amit

```
In [100... Dict11 = {'Amit': [70, 80, 90], 'Riya': [85, 75, 80], 'Neha': [60, 70, 65]}
Topper = ""
maximum = 0
for i,j in Dict11.items():
    if maximum < sum(j):
        maximum = sum(j)
        Topper = i
print("Topper:", Topper)
```

Topper: Amit

12) WAP to extract keys whose values are even numbers.

Sample Input:

```
{1: 11, 2: 20, 3: 15, 4: 40}
```

Sample Output:

[2, 4]

```
In [94]: Dict12 = {1: 11, 2: 20, 3: 15, 4: 40}
newDict12= {}
for i,j in Dict12.items():
    if j % 2 == 0:
        newDict12[i] = j
li12 = newDict12.keys()
print(li12)
```

dict_keys([2, 4])

13) WAP to Find common keys between two dictionaries.

Sample Input:

```
Dict1 = {1: 'A', 2: 'B', 3: 'C'}
Dict2 = {2: 'X', 3: 'Y', 4: 'Z'}
```

Sample Output:

[2, 3]

```
In [104... Dict13 = {1: 'A', 2: 'B', 3: 'C'}
Dict13_2 = {2: 'X', 3: 'Y', 4: 'Z'}
New_Dict = []
for i,j in Dict13.items():
    if i in
print(New_Dict)
```

dict_keys([2, 3, 4])

14) WAP to handle missing keys in dictionaries.

Sample Input - 1:

```
dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'c'
```

Sample Output - 1:

8

Sample Input - 2:

```
dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'd'
```

Sample Output - 2:

Key Not Found

```
In [2]: dict14 = {'a': 5, 'c': 8, 'e': 2}
key = 'c'
key2 = 'd'
if key in dict14:
    print(dict14[key])
else:
    print("Key not found")

if key2 in dict14:
    print(dict14[key2])
else:
    print("Key not found")
```

8

Key not found

15) WAP to create a dictionary using dictionary comprehension from two lists containing student names and marks. Include only those students in the dictionary whose marks are greater than or equal to 75.

Sample Input:

```
students = ['Amit', 'Riya', 'Neha', 'Karan']  
marks = [78, 72, 85, 60]
```

Sample Output:

```
{'Amit': 78, 'Neha': 85}
```

```
In [14]: students = ['Amit', 'Riya', 'Neha', 'Karan']  
marks = [78, 72, 85, 60]  
  
dict15 = {students[i]:marks[i] for i in range(0,len(students),1) if marks[i] >=  
print(dict15)  
  
{'Amit': 78, 'Neha': 85}
```

```
In [ ]:
```